

Chulhong Min

PRINCIPAL RESEARCH SCIENTIST AND TECH LEAD · NOKIA BELL LABS, CAMBRIDGE, UK

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Research Profile

I am a Principal Research Scientist and Tech Lead with leadership experience in industry-driven AI R&D, specializing in the efficient deployment of ML and LLMs on resource-constrained devices. I lead the Device Systems team in building scalable multi-device AI systems across wearables, IoT, and edge devices. I have expertise in on-device AI, adaptive and distributed inference, and edge AI accelerators, enabling scalable, low-power AI services. I am passionate about transforming advanced research into deployable products and impactful applications.

Professional Experience

- 2022-NOW **Principal Research Scientist and Tech Lead**, Nokia Bell Labs
- 2019-2022 **Research Scientist**, Nokia Bell Labs
- 2019-2021 **Visitor**, University of Cambridge
- 2017-2019 **Member of Technical Staff**, Nokia Bell Labs
- 2016 **Postdoctoral Researcher**, KAIST

Education

KAIST (Korea Advanced Institute of Science and Technology)

Daejeon, South Korea

PH.D. IN COMPUTER SCIENCE

Feb. 2016

- Advisor: Dr. June-hwa Song
- [Outstanding Ph.D Thesis Award](#) in School of Computing

KAIST (Korea Advanced Institute of Science and Technology)

Daejeon, South Korea

M.S. IN COMPUTER SCIENCE

Aug. 2009

KAIST (Korea Advanced Institute of Science and Technology)

Daejeon, South Korea

B.S. IN COMPUTER SCIENCE

Aug. 2007

Research Experience

Multi-modal Sensory LLM Frameworks for Edge/Cloud Collaborative Deployment

Nokia Bell Labs

2025-

- I lead an initiative on extending Large Language Models (LLMs) to multi-modal sensory data from wearables, tackling the challenge of deploying such pipelines on resource-constrained devices. My team is characterizing their resource and operational demands and developing a modular framework that enables scalable, distributed deployment via edge-cloud collaboration.

AI-native Multi-sensory Programmable Runtime for Ultra-low Power Devices

Nokia Bell Labs

BLUEPRINT, BLOG, PAPERS ([J.17], [C.22], [W.17], [W.15], [W.14], [W.12])

2023-

- I lead the development of an AI-native, ultra-efficient runtime for multi-sensory wearables with tiny AI accelerators. The project advances heterogeneous AI accelerator orchestration, holistic storage management, and software-defined thermal management, thereby enabling seamless AI services on intelligent, screenless, and formless devices.

On-the-fly Device-to-device Authentication

Nokia Bell Labs

PAPERS ([C.23], [J.15], [W.16])

2023-

- Today's tiny-form wearables lack biometric hardware or touchscreens, often relying on one-time Bluetooth pairing. My team develops secure, adaptable authentication strategies to enable dynamic collaboration among screenless on-body devices.

Bringing MLOPs and Multi-tenant Model Serving to Sensory Edge Devices

Nokia Bell Labs

PAPERS ([J.16], [J.14], [J.13], [J.11], [J.10], [C.18], [C.16], [W.07])

2021-2024

- I led the development of a multi-tenant runtime for adaptive model execution with integrated MLOps on edge devices, which leverages redundant sensory signals and distributed computing resources. I guided the project from research to industry application, rebranded **Camera-as-a-Service**, and integrated into smart camera platforms to deliver scalable video analytics.

Camera-as-a-Service at U23 and Slush

Nokia Bell Labs

PROJECT, VIDEO, NOKIA CEO'S MESSAGE, NOKIA BELL LABS PRESIDENT'S BLOG

2022-2023

- *European Athletics U23 Championships*: I led the end-to-end deployment of our Camera-as-a-Service solution at the 2023 European Athletics U23 Championships in Espoo, Finland. I coordinated cross-functional efforts with the U23 organizing committee, venue managers, subcontractors, and Nokia's legal team to ensure successful delivery. The deployment provided thousands of spectators with unique digital services, showcasing the real-world scalability and impact of our solution.
- *Demonstration at Slush*: In 2022, I demonstrated our Camera-as-a-Service solution to a global audience at Slush, one of the world's leading startup and tech events in Helsinki, Finland, with more than 12,000 attendees.

eSense - Open Earable Platform for Human Sensing

Nokia Bell Labs

PROJECT, PAPERS ([J.09], [C.14], [C.13], [J.05], [W.05], [W.04], [D.10], [D.09], [D.08], [P.03])

2017-2019

- I contributed to the eSense project, developing the world's first open earable platform for human sensing. eSense integrates audio, motion, and BLE sensing for behavior analytics, cognitive assistance, and real-time well-being monitoring. We distributed 1,000 eSense prototypes to over 70 research labs worldwide, playing a key role in pioneering early earable research.

Mobile, Wearable, and Embedded Systems for Life-immersive Applications

KAIST

PAPERS ([J.08], [C.10], [J.04], [J.03], [C.09], [C.08], [C.07], [J.02], [C.05], [C.04], [C.03], [J.01], [C.02], [C.01], [T.02], [T.01], [D.06], [D.05], [D.04], [P.02], [D.03], [P.01], [D.02], [F.01], [I.02], [D.01], [I.01])

2007-2017

- During my PhD, I developed software systems to support life-immersive applications on mobile, wearable, and embedded platforms. I addressed system challenges of continuous human sensing on resource-constrained devices through innovations in novel sensing, resource management, device collaboration, and user perception, devising novel systems solutions.

Mobile, Social, and Earable Applications

Nokia Bell Labs, KAIST

PAPERS ([C.21], [C.20], [J.07], [C.15], [C.11], [C.06], [W.13], [W.10], [W.09], [W.06], [W.02], [W.01], [P.07], [P.06], [P.05], [P.05], [D.07], [V.01])

2007-

- As a systems researcher, I enjoyed developing innovative mobile, social, and earable applications on top of the systems I built. This hands-on work inspired new design insights and revealed novel requirements for next generation systems.

Datasets

Nokia Bell Labs, KAIST

SMARTWATCH [C.07], AMBIENT ACOUSTIC [C.17], FATIGUESet [C.19], WEEE [J.13]

2007-

- To provide valuable insights for further exploration, I have released multiple datasets collected during my studies.

Publications

2025

- [J.17] Taesik Gong, SiYoung Jang, Utku Günay Acer, Fahim Kawsar, **Chulhong Min**, *Synergy: Towards On-Body AI via Tiny AI Accelerator Collaboration on Wearables*, In IEEE Transactions on Mobile Computing. In IEEE Transaction on Mobile Computing (TMC), Volume 24, Issue: 10, pp 9319-9333.
- [C.23] Adiba Orzikulova, Diana A. Vasile, Chi Ian Tang, Fahim Kawsar, Sung-ju Lee, **Chulhong Min**, *BioQ: Towards Context-Aware Multi-Device Collaboration with Bio-cues*, In Proc. ACM Conference on Embedded Networked Sensor Systems (ACM SenSys 2025), Irvine, USA, May 2025
- [W.17] Changmin Jeon, Taesik Gong, Juheon Yi, Fahim Kawsar, **Chulhong Min**, *TinyMem: Boosting Multi-DNN Inference on Tiny AI Accelerators with Weight Memory Virtualization*, In Proc. International Workshop on Mobile Computing Systems and Applications (ACM HotMobile 2025), Palm Springs, California, USA, February 2025.
- [W.16] Sujin Han, Diana A. Vasile, Fahim Kawsar, **Chulhong Min**, *SecuWear: Secure Data Sharing Between Wearable Devices*, In Proc. Workshop on Security and Privacy in Standardized IoT (SDIoTSec 2025) in conjunction with NDSS 2025, San Diego, California, February 2025.

2024

- [C.22] Taesik Gong, Fahim Kawsar, **Chulhong Min**, *DEX: Data Channel Extension for Efficient CNN Inference on Tiny AI Accelerators*, In Proc. Thirty-Eighth Annual Conference on Neural Information Processing Systems (NeurIPS 2024), Vancouver, Canada, December 2024.
- [W.15] Yushan Huang, Taesik Gong, SiYoung Jang, Fahim Kawsar, **Chulhong Min**, *Energy Characterization of Tiny AI Accelerator-Equipped Microcontrollers*, In Proc. International Workshop on Human-centered Sensing, Networking, and Multi-Device Systems (HumanSys 2024) in conjunction with ACM SenSys 2024, Hangzhou, China, November 2024. [\[Best paper award\]](#)
- [J.16] Juheon Yi, Utku Gunay Acer, Fahim Kawsar, **Chulhong Min**, *Argus: Enabling Cross-Camera Collaboration for Video Analytics on Distributed Smart Cameras*, In IEEE Transaction on Mobile Computing (TMC), Volume 24, Issue: 1, pp 117-134.
- [J.15] Diana A. Vasile, Fahim Kawsar, **Chulhong Min**, *Emerging Paradigms in Wearable Security: Adaptable and Secure Sandboxing for On-the-Fly Collaboration Among Wearables*, In IEEE Security & Privacy. Volume: 22, Issue: 6, pp 30-38.
- [W.14] SiYoung Jang, Fahim Kawsar, **Chulhong Min**, *Thermal Characterization of AI Applications on AI Accelerators-equipped Microcontrollers*, In Proc. International Workshop on Body-Centric Computing Systems (BodySys 2024) in conjunction with ACM MobiSys 2024, Tokyo, Japan, June 2024.
- [A.03] **Chulhong Min**, Utku Günay Acer, Siyoung Jang, Sangwon Choi, Diana A. Vasile, Taesik Gong, Juheon Yi, Fahim Kawsar, *An AI-Native Runtime for Multi-Wearable Environments*, arXiv preprint arXiv:2403.17863 (2024)
- [A.02] Adiba Orzikulova, Diana A. Vasile, Fahim Kawsar, **Chulhong Min**, *Time-bound Contextual Bio-ID Generation for Minimalist Wearables*, arXiv preprint arXiv:2403.00889 (2024)

2023

- [C.21] Euihyeok Lee, **Chulhong Min**, Jaeseung Lee, Jin Yu, Seungwoo Kang, *GrooveMeter: Enabling Music Engagement-aware Apps by Detecting Reactions to Daily Music Listening via Earable Sensing*, In Proc. ACM International Conference on Multimedia (MM), Ottawa, Canada, Oct. 2023.
- [J.14] **Chulhong Min**, Akhil Mathur, Utku Günay Acer, Alessandro Montanari, Fahim Kawsar, *SensiX++: Bringing MLOPs and Multi-tenant Model Serving to Sensory Edge Devices*, In ACM Transactions on Embedded Computing Systems (TECS), Volume 22, Issue 6, Article No.: 98, pp 1-27.
- [J.13] **Chulhong Min**, Akhil Mathur, Alessandro Montanari, Fahim Kawsar, *SensiX: A System for Best-effort Inference of Machine Learning Models in Multi-device Environments*, In IEEE Transactions on Mobile Computing (TMC), Volume 22, Issue 9, pp 5525-5538.
- [W.13] Bhawana Chhagani, Utku Günay Acer, SiYoung Jang, Fahim Kawsar, **Chulhong Min**, *Cocoon: On-body Microphone Collaboration for Spatial Awareness*, In Proc. International Workshop on Mobile Computing Systems and Applications (ACM HotMobile 2023), Orange County, California, USA, Feb 2023.

2022

- [W.12] Arthur Moss, Hyunjong Lee, Lei Xun, **Chulhong Min**, Fahim Kawsar, Alessandro Montanari, *Ultra-low Power DNN Accelerators for IoT: Resource Characterisation of the MAX78000*, In Proc. ACM Workshop on Challenges in Artificial Intelligence and Machine Learning for Internet of Things (AIChallengIoT 2022) in conjunction with ACM SenSys 2022, Boston, USA, November 2022. [\[Best paper award\]](#)
- [J.12] Shkurta Gashi, **Chulhong Min**, Alessandro Montanari, Silvia Santini, Fahim Kawsar, *A Multi-device and Multi-modal Dataset for Human Energy Expenditure Estimation using Wearable Devices*, In Nature Scientific Data 9, Article number 537 (2022).
- [J.11] Utku Günay Acer, Marc Van Den Broeck, **Chulhong Min**, Mallesh Dasari, Fahim Kawsar, *The City as a Personal Assistant: Turning Urban Landmarks into Conversational Agents for Serving Hyper Local Information*, In Proc. ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (ACM IMWUT). Volume 6, Issue 2, Article 40 (July 2022), 31 pages.
- [J.10] Yash Jain, Chi Ian Tang, **Chulhong Min**, Fahim Kawsar, Akhil Mathur, *ColloSSL: Collaborative Self-Supervised Learning for Human Activity Recognition*, In Proc. ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (ACM IMWUT). Volume 6, Issue 1, Article 17 (March 2022), 28 pages.
- [C.20] Jing Yang, **Chulhong Min**, Akhil Mathur, Fahim Kawsar, *SleepGAN: Towards Personalized Sleep Therapy Music*, In Proc. 2022 IEEE International Conference on Acoustics, Speech and Signal Processing (IEEE ICASSP 2022), Singapore, May 2022.

[J.09] Andrea Ferlini, Alessandro Montanari, **Chulhong Min**, Hongwei Li, Ugo Sassi, Fahim Kawsar, *In-Ear PPG for Vital Signs*, IEEE Pervasive Computing, Volume 21, Issue 1, p. 65-74, Jan.-March 2022.

2021

[C.19] Manasa Kalanadhabhatta, **Chulhong Min**, Alessandro Montanari, Fahim Kawsar, *FatigueSet: A Multi-modal Dataset for Modeling Mental Fatigue and Fatigability*, In Proc. EAI International Conference on Pervasive Computing Technology for Healthcare (PervasiveHealth 2021), December 2021.

[C.18] SiYoung Jang, Utku Günay Acer, **Chulhong Min**, Fahim Kawsar, *Deploying Collaborative Machine Learning Systems in Edge with Multiple Cameras*, In Proc. the 13th International Conference on Mobile Computing and Ubiquitous Networking (ICMU 2021), November 2021. [\[Best student paper award\]](#)

[W.11] Juheon Yi, **Chulhong Min**, Fahim Kawsar, *Vision Paper: Towards Software-Defined Video Analytics with Cross-Camera Collaboration*, In Proc. ACM Workshop on Challenges in Artificial Intelligence and Machine Learning for Internet of Things (AIChallengerIoT 2021) in conjunction with ACM SenSys 2021, Coimbra, Portugal, November 2021.

[W.10] Dongwoo Kim, **Chulhong Min**, Seungwoo Kang, *Towards Automatic Recognition of Perceived Level of Understanding on Online Lectures using Earables*, In Proc. ACM Workshop on Earable Computing (ACM EarComp 2021) in conjunction with ACM UbiComp 2021, Virtual, September 2021.

[W.09] Matija Franklin, David Lagnado, **Chulhong Min**, Akhil Mathur, Fahim Kawsar, *Designing Memory Aids for Dementia Patients using Earables*, In Proc. ACM Workshop on Earable Computing (ACM EarComp 2021) in conjunction with ACM UbiComp 2021, Virtual, September 2021

[A.01] **Chulhong Min**, Akhil Mathur, Utku Günay Acer, Alessandro Montanari, Fahim Kawsar *SensiX++: Bringing MLOPs and Multi-tenant Model Serving to Sensory Edge Devices*, arXiv preprint arXiv:2109.03947 (2019)

[W.08] Yash Jain, Chi Ian Tang, **Chulhong Min**, Fahim Kawsar, Akhil Mathur, *Group Supervised Learning: Extending Self-Supervised Learning to Multi-Device Settings*, In Proc. ICML 2021 Workshop Self-Supervised Learning for Reasoning and Perception in conjunction with ICML 2021, Virtual, July 2021.

2020

[P.07] Dongwoo Kim, **Chulhong Min**, Seungwoo Kang, *Poster Abstract: Towards Recognizing Perceived Level of Understanding for Online Lectures using Earables*, In Proc. ACM Conference on Embedded Networked Sensor Systems (ACM SenSys 2020), Yokohama, Japan, November 2020.

[P.06] Euihyeok Lee, Dongwoo Kim, **Chulhong Min**, Seungwoo Kang, *Poster Abstract: Automatic Recognition of Vocal Reactions in Music Listening using Smart Earbuds*, In Proc. ACM Conference on Embedded Networked Sensor Systems (ACM SenSys 2020), Yokohama, Japan, November 2020.

[C.17] Chunjong Park, **Chulhong Min**, Sourav Bhattacharya, Fahim Kawsar, *Augmenting Conversational Agents with Ambient Acoustic Contexts*, In Proc. the 22nd International Conference on Human-Computer Interaction with Mobile Devices and Services (ACM MobileHCI 2020), October 2020.

[J.08] **Chulhong Min**, Youngki Lee, Chungkuk Yoo, Inseok Hwang, Younghyun Ju, Junehwa Song, Seungwoo Kang, *Scalable Power Impact Prediction of Mobile Sensing Applications at Pre-installation Time*, In IEEE Transactions on Mobile Computing (TMC), Volume 19, Issue 6, pp. 1448-1464, June 2020.

2019

[C.16] **Chulhong Min**, Alessandro Montanari, Akhil Mathur, Fahim Kawsar, *A Closer Look at Quality-Aware Runtime Assessment of Sensing Models in Multi-Device Environments*, In Proc. the 17th ACM Conference on Embedded Networked Sensor Systems (ACM SenSys 2019), New York, USA, Nov. 10-13, 2019.

[W.07] Mattia Antonini, Tran Huy Vu, **Chulhong Min**, Alessandro Montanari, Akhil Mathur, Fahim Kawsar, *Resource Characterisation of Personal-Scale Sensing Models on Edge Accelerators*, In Proc. ACM Workshop on Challenges in Artificial Intelligence and Machine Learning for Internet of Things (AIChallengerIoT 2019) in conjunction with ACM SenSys 2019, New York, USA, November 2019.

[J.07] Inseok Hwang, Youngki Lee, Chungkuk Yoo, **Chulhong Min**, Dongsun Yim, John Kim, *Towards Interpersonal Assistants: Next-generation Conversational Agents*, IEEE Pervasive Computing, Volume 18, Issue 2, p. 21-31, August 2019.

[C.15] **Chulhong Min**, Euihyeok Lee, Souneil Park, Seungwoo Kang, *Tiger: Wearable Glasses for the 20-20-20 Rule to Alleviate Computer Vision Syndrome*, In Proc. 21st International Conference on Human-Computer Interaction with Mobile Devices and Services (MobileHCI 2019), Taipei, Taiwan, Oct. 1-4, 2019. [\[Honorable Mention Award\]](#)

- [C.14] **Chulhong Min**, Akhil Mathur, Alessandro Montanari, Fahim Kawsar, *An Early Characterisation of Wearing Variability on Motion Signals for Wearables*, In Proc. the 2019 International Symposium on Wearable Computers (ISWC 2019), London, UK, Sep. 11-13, 2019.
- [W.06] Chungkuk Yoo, Seungwoo Kang, Inseok Hwang, **Chulhong Min**, Seonghoon Kim, Wonjung Kim, Junehwa Song, *Mom, I see You Angry at Me! Designing a Mobile Service for Parent-child Conflicts by In-situ Emotional Empathy*, In Proc. ACM Workshop on Mobile Systems for Computational Social Science (ACM MCSS 2019) in conjunction with ACM MobiSys 2019, Seoul, South Korea, June 2019.
- [C.13] Seungchul Lee, **Chulhong Min**, Alessandro Montanari, Akhil Mathur, Youngjae Chang, Junehwa Song, Fahim Kawsar, *Automatic Smile and Frown Recognition with Kinetic Earables*, In Proc. 10th Augmented Human International Conference (AH 2019), Reims Champagne-Ardenne, France, March 2019.
- [J.06] Mohammed Alloulah, Anton Isopoussu, **Chulhong Min**, Fahim Kawsar, *On Tracking the Physicality of Wi-Fi: A Subspace Approach*, IEEE Access, Volume 7, p. 19965-19978, February 2019.

2018

- [C.12] **Chulhong Min**, Mohammed Alloulah, Fahim Kawsar, *An Early Resource Characterisation of Wi-Fi Sensing on Residential Gateways*, In Proc. ACM International Conference on Energy-Efficient Built Environments (ACM BuildSys 2018), Shenzhen, China, November 2018.
- [D.10] Fahim Kawsar, **Chulhong Min**, Akhil Mathur, Alessandro Montanari, Utku Günay Acer, Marc Van den Broeck, *Demo Abstract: eSense - Open Earable Platform for Human Sensing*, In Proc. ACM Conference on Embedded Networked Sensor Systems (ACM SenSys 2018), Shenzhen, China, November 2018.
- [J.05] Fahim Kawsar, **Chulhong Min**, Akhil Mathur, Alessandro Montanari, *Earables for Personal-Scale Behavior Analytics*, IEEE Pervasive Computing, Volume 17, Issue 3, p. 83-89, October 2018
- [W.05] **Chulhong Min**, Alessandro Montanari, Akhil Mathur, Seungchul Lee, Fahim Kawsar, *Cross-Modal Approach for Conversational Well-being Monitoring with Multi-Sensory Earables*, In Proc. ACM Workshop on Computing for Well-being (ACM WellComp 2018) in conjunction with ACM UbiComp 2018, Singapore, Singapore, October 2018.
- [D.09] Fahim Kawsar, **Chulhong Min**, Akhil Mathur, Alessandro Montanari, Utku Günay Acer, Marc Van den Broeck, *Demo: eSense - Open Earable Platform for Human Sensing*, In Proc. ACM International Joint Conference and International Symposium on Pervasive and Ubiquitous Computing and Wearable Computers (ACM UbiComp 2018), Singapore, Singapore, October 2018.
- [P.05] Euihyeok Lee, **Chulhong Min**, Seungwoo Kang, *Poster: Towards a Wearable Assistant to Prevent Computer Vision Syndrome*, In Proc. ACM International Joint Conference and International Symposium on Pervasive and Ubiquitous Computing and Wearable Computers (ACM UbiComp 2018), Singapore, Singapore, October 2018.
- [P.04] Seungchul Lee, Saumay Pushp, **Chulhong Min**, Junehwa Song, *Poster: Exploring Relationship-aware Dynamic Message Screening for Mobile Messengers*, In Proc. ACM International Joint Conference and International Symposium on Pervasive and Ubiquitous Computing and Wearable Computers (ACM UbiComp 2018), Singapore, Singapore, October 2018.
- [W.04] **Chulhong Min**, Akhil Mathur, Fahim Kawsar, *Exploring Audio and Kinetic Sensing on Earable Devices*, In Proc. ACM Workshop on Wearable Systems and Applications (ACM WearSys 2018) in conjunction with ACM MobiSys 2018, Munich, Germany, June 2018.
- [D.08] Fahim Kawsar, **Chulhong Min**, Akhil Mathur, Marc Van den Broeck, Utku Günay Acer, Claudio Forlivesi, *Demo: eSense - Open Earable Platform for Human Sensing*, In Proc. ACM Annual International Conference on Mobile Systems, Applications, and Services (ACM MobiSys 2018), Munich, Germany, June 2018.
- [P.03] **Chulhong Min**, Akhil Mathur, Fahim Kawsar, *Poster: Audio-Kinetic Model for Automatic Dietary Monitoring with Earable Devices*, In Proc. ACM Annual International Conference on Mobile Systems, Applications, and Services (ACM MobiSys 2018), Munich, Germany, June 2018.

2017

- [C.11] Bumsoo Kang, **Chulhong Min**, Wonjung Kim, Inseok Hwang, Chunjong Park, Seungchul Lee, Sung-ju Lee, Junehwa Song, *Zaturi: We Put Together the 25th Hour for You. Create a Book for Your Baby*, In Proc. ACM Conference on Computer-Supported Cooperative Work and Social Computing (ACM CSCW 2017), Portland, Oregon, USA, February 2017.
- [D.07] Bumsoo Kang, Wonjung Kim, Inseok Hwang, Chunjong Park, Seungchul Lee, **Chulhong Min**, Sung-Ju Lee, Junehwa Song, *Zaturi: Blending Hours Spent at Work and Hours Devoted to Children*, In Proc. ACM Conference on Computer-Supported Cooperative Work and Social Computing (ACM CSCW 2017), Portland, Oregon, USA, February 2017.

2016

- [C.10] **Chulhong Min**, Seungchul Lee, Changhun Lee, Youngki Lee, Seungwoo Kang, Seungpyo Choi, Wonjung Kim, June-hwa Song, *PADA: Power-aware Development Assistant for Mobile Sensing Applications*, In Proc. ACM International Joint Conference on Pervasive and Ubiquitous Computing (ACM UbiComp 2016), Heidelberg, Germany, September 2016.
- [J.04] Youngki Lee, Seungwoo Kang, **Chulhong Min**, Younghyun Ju, Inseok Hwang, Junehwa Song, *CoMon+: A Cooperative Context Monitoring System for Multi-Device Personal Sensing Environments*, In IEEE Transactions on Mobile Computing (TMC), Volume 15, Issue 8, pp. 1908-1924, August 2016.
- [J.03] **Chulhong Min**, Youngki Lee, Chungkuk Yoo, Seungwoo Kang, Inseok Hwang, Junehwa Song, *PowerForecaster: Predicting Power Impact of Mobile Sensing Applications at Pre-installation Time*, In GetMobile: Mobile Comp. and Comm. Volume 20, Issue 1 (July 2016), pages 30-33.
- [T.01] **Chulhong Min**, *User Support for Battery Management of Continuous Sensing Applications*, Ph.D. Thesis, School of Computing, KAIST, Feb 2016. [\[Outstanding Ph.D. Thesis Award\]](#)

2015

- [C.09] **Chulhong Min**, Youngki Lee, Chungkuk Yoo, Seungwoo Kang, Sangwon Choi, Pillsoon Park, Inseok Hwang, Younghyun Ju, Seungpyo Choi, Junehwa Song, *PowerForecaster: Predicting Smartphone Power Impact of Continuous Sensing Applications at Pre-installation Time*, In Proc. ACM Conference on Embedded Networked Sensor Systems (ACM SenSys 2015), Seoul, Korea, November 2015.
- [D.06] **Chulhong Min**, Chungkuk Yoo, Sangwon Choi, Pillsoon Park, Seungchul Lee, Changhun Lee, Seungpyo Choi, Seungwoo Kang, Youngki Lee, Inseok Hwang, Younghyun Ju, Junehwa Song, *Demo: User Support for Power Management of Continuous Sensing Applications*, In Proc. ACM Conference on Embedded Networked Sensor Systems (ACM SenSys 2015), Seoul, Korea, November 2015.
- [C.08] **Chulhong Min**, Chungkuk Yoo, Inseok Hwang, Seungwoo Kang, Youngki Lee, Seungchul Lee, Pillsoon Park, Changhun Lee, Seungpyo Choi, Junehwa Song, *Sandra Helps You Learn: the More You Walk, the More Battery Your Phone Drains*, In Proc. ACM International Joint Conference on Pervasive and Ubiquitous Computing (ACM UbiComp 2015), Osaka, Japan, September 2015.
- [C.07] **Chulhong Min**, Seungwoo Kang, Chungkuk Yoo, Jeehoon Cha, Sangwon Choi, Younghun Oh, Junehwa Song, *Exploring Current Practices for Battery Use and Management of Smartwatches*, In Proc. Proceedings of the 19th International Symposium on Wearable Computers (ACM ISWC 2015), Osaka, Japan, September 2015.
- [W.03] Sebastian Deterding, Andrés Lucero, Jussi Holopainen, **Chulhong Min**, Adrian Cheok, Annika Waern, Steffen Walz, *Embarrassing Interactions*, In Embarrassing Interactions 2015, in conjunction with ACM CHI 2015, Seoul, Korea, April 2015.

2014

- [W.02] **Chulhong Min**, Saumay Pushp, Seungchul Lee, Inseok Hwang, Youngki Lee, Seungwoo Kang, Junehwa Song, *Uncovering Embarrassing Moments in In-situ Exposure of Incoming Mobile Messages*, In Proc. ACM Workshop on Mobile Systems for Computational Social Science (ACM MCSS 2014) in conjunction with ACM UbiComp 2014, Seattle, USA, September 2014.
- [J.02] Youngki Lee, **Chulhong Min**, Younghyun Ju, Seungwoo Kang, Yunseok Rhee, Junehwa Song, *An Active Resource Orchestration Framework for PAN-Scale, Sensor-Rich Environments*, In IEEE Transactions on Mobile Computing (TMC), Vol. 13, Issue 3, March 2014.
- [C.06] Inseok Hwang, Chungkuk Yoo, Chanyou Hwang, Dongsun Yim, Youngki Lee, **Chulhong Min**, John Kim, Junehwa Song, *TalkBetter: family-driven mobile intervention care for children with language delay*, In Proc. ACM conference on Computer supported cooperative work and social computing (ACM CSCW 2014), Baltimore, USA, February 2014. [\[Best paper award\]](#)

2013

- [W.01] SangJeong Lee, **Chulhong Min**, Chungkuk Yoo, Junehwa Song, *Understanding Customer Mallling Behavior in an Urban Shopping Mall using Smartphones*, In Proc. ACM Workshop on Mobile Systems for Computational Social Science (ACM MCSS 2013) in conjunction with ACM UbiComp 2013, Zurich, Switzerland, September 2013.
- [C.05] Youngki Lee, **Chulhong Min**, Chanyou Hwang, Jaeung Lee, Inseok Hwang, Younghyun Ju, Chungkuk Yoo, Miri Moon, Uichin Lee, Junehwa Song, *SocioPhone: Everyday Face-to-Face Interaction Monitoring Platform using Multi-phone Sensor Fusion*, In Proc. ACM Annual International Conference on Mobile systems, applications, and services (ACM MobiSys 2013), Taipei, Taiwan, June 2013.

[D.05] Youngki Lee, **Chulhong Min**, Chanyou Hwang, Jaeung Lee, Inseok Hwang, Younghyun Ju, Chungkuk Yoo, Miri Moon, Uichin Lee, Junehwa Song, *Demo - SocioPhone: Everyday Face-to-Face Interaction Monitoring Platform using Multi-phone Sensor Fusion*, In Proc. ACM Annual International conference on Mobile systems, applications, and services (ACM MobiSys 2013), Taipei, Taiwan, June 2013.

[D.04] **Chulhong Min**, Inseok Hwang, Jaeung Lee, Chanyou Hwang, Chungkuk Yoo, Miri Moon, Taiwoo Park, Changhoon Lee, Haechan Lee, Yuhwan Kim, Younghyun Ju, Youngki Lee, Uichin Lee, Junehwa Song, *ACM HotMobile 2013 Demo: Brining In-situ Social Awareness to Mobile Systems: Conversational Turn Monitoring and its Applications*, In MC2R '13: ACM Mobile Computing and Communication Review, vol. 17, no. 3 (2013) and In ACM Workshop on Mobile Computing Systems and Applications (ACM HotMobile 2013), Jekyll Island, USA, February 2013. [\[Best Demo Award\]](#)

2012

[C.04] Younghyun Ju, Youngki Lee, Jihyun Yu, **Chulhong Min**, Insik Shin, Junehwa Song, *SymPhoney: A Coordinated Sensing Flow Execution Engine for Concurrent Mobile Sensing Applications*, In Proc. ACM Conference on Embedded Network Sensor Systems (ACM SenSys 2012), Toronto, Canada, November 2012.

[P.02] Saumay Pushp, **Chulhong Min**, Youngki Lee, Chi Harold Lie, Junehwa Song, *Towards Crowd-aware Sensing Platform for Metropolitan Environments*, In Proc. ACM Conference on Embedded Network Sensor Systems (ACM SenSys 2012), Toronto, Canada, November 2012.

[C.03] Youngki Lee, Younghyun Ju, **Chulhong Min**, Seungwoo Kang, Inseok Hwang, Junehwa Song, *CoMon: Cooperative Ambience Monitoring Platform with Continuity and Benefit Awareness*, In Proc. ACM International Conference on Mobile systems, applications, and services (ACM MobiSys 2012), Low Wood Bay, Lake District, United Kingdom, June 2012.

[P.01] **Chulhong Min**, Wookhyun Han, Inseok Hwang, SangJeong Lee, Youngki Lee, Insik Shin, Junehwa Song, *Poster: Towards Mobile GPU-Accelerated Context Processing for Continuous Sensing Applications on Smartphones*, In Proc. Annual International Conference on Mobile systems, applications, and services (ACM MobiSys 2012), Low Wood Bay, Lake District, United Kingdom, June 2012.

[F.01] **Chulhong Min**, *SensorShader: Mobile GPU-Accelerated Context Processing Engine for Sensing Applications on Smartphones*, In MobiSys Ph.D. Forum, Low Wood Bay, Lake District, United Kingdom, June 2012.

[D.03] Youngki Lee, Younghyun Ju, **Chulhong Min**, Jihyun Yu, Junehwa Song, *MobiCon: Mobile context monitoring platform, Incorporating Context-awareness to Smartphone-centric Personal Sensor Networks*, In IEEE Communications Society Conference on Sensor, Mesh and Ad Hoc Communications and Networks (IEEE SECON 2012), Seoul, Korea, June 2012.

[D.02] Youngki Lee, Younghyun Ju, **Chulhong Min**, Seungwoo Kang, Inseok Hwang, Junehwa Song, *Demo: SenseTogether-Cooperative Ambience Monitoring Platform with Continuity and Benefit Awareness*, In Proc. Annual International Conference on Mobile systems, applications, and services (ACM MobiSys 2012), Low Wood Bay, Lake District, United Kingdom, June 2012.

[J.01] Youngki Lee, Sitharam S. Iyengar, **Chulhong Min**, Younghyun Ju, Seungwoo Kang, Taiwoo Park, Jinwon Lee, Yunseok Rhee, Junehwa Song, *MobiCon: A Mobile Context-monitoring Platform*, In Communications of the ACM (CACM), Vol. 55, Issue 3, March 2012.

[C.02] Younghyun Ju, **Chulhong Min**, Youngki Lee, Jihyun Yu, Junehwa Song, *An Efficient Dataflow Execution Method for Mobile Context Monitoring Applications*, In Proc. IEEE International Conference on Pervasive Computing and Communications (PerCom 2012), Lugano, Switzerland, March 2012.

2011

[I.02] **Chulhong Min**, Chungkuk Yoo, Youngki Lee, Junehwa Song, *Healthopia: Towards Your Well-Being in Everyday Life (invited paper)*, In Proc. International Symposium on Applied Sciences in Biomedical and Communication Technologies (ISABEL 2011), Barcelona, Spain, October 2011.

[D.01] Youngki Lee, Younghyun Ju, **Chulhong Min**, Seungwoo Kang, Yunseok Rhee, Junehwa Song, *Demo: CoMon – Resource-aware Cooperative Context Monitoring System for Smartphone-centric Sensor-rich PANs*, In Proc. ACM Annual International Conference on Mobile systems, applications, and services (ACM MobiSys 2011), Washington, DC, USA, June 2011.

[I.01] Youngki Lee, **Chulhong Min**, Younghyun Ju, Saumay Pushp, Junehwa Song, *A Mobile Context Monitoring Platform for Pervasive Computing Environments (Invited paper)*, In IEEE International Conference on Digital Ecosystems and Technologies (IEEE DEST 2011), Daejeon, Korea, May 2011.

2010

[C.01] Seungwoo Kang, Youngki Lee, **Chulhong Min**, Younghyun Ju, Taiwoo Park, Jinwon Lee, Yunseok Rhee, Junehwa Song, *Orchestrator: An Active Resource Orchestration Framework for Mobile Context Monitoring in Sensor-rich Mobile*

Patents

GRANTED

Personalized models (GB2588689A)

Detection of facial expressions (US12299187B2, CN113557490B, WO2020182447A1, EP3709131A1 - pending)

System and method for automatic detection of music listening reactions, and mobile device performing the method (US20230171541A1, KR102735260B1)

Presence or activity detection (EP3459439)

Method and devices for processing sensor data by applying one or more processing pipelines to the sensor data (US11557898, WO2019091959A1)

Communication device for predicting power consumption of mobile application, communication system including same, method of predicting power consumption of mobile application and method of providing predicted power consumption of mobile application, using same (US10928877B2, WO2017155322A2, KR101758267B1)

Sensing in wireless communication systems (EP3624410)

Mobile apparatus, audio book creating system having the same and method of creating audio book using the same (KR102061619B1)

Language delay treatment system and control method for the same (US9875668B2)

Mobile device executing face-to-face interaction monitoring, method of monitoring face-to-face interaction using the same, and interaction monitoring system including the same, and mobile interaction monitoring application executed on the same (US9813879B2)

Mobile device for performing monitoring on cooperative situation, method of performing monitoring on cooperative situation using said mobile device, and system for performing monitoring on cooperative situation including said mobile device (US9756095B2, WO2014046369A1 - pending)

Mobile apparatus supporting cooperative context monitoring, method of cooperative context monitoring using the same and cooperative context monitoring system including the same (US9756095B2)

Mobile apparatus for executing sensing flow for mobile context monitoring, method of executing sensing flow using the same, method of context monitoring using the same and context monitoring system including the same (US9367664B2, KR101549002B1)

Mobile apparatus supporting context monitoring, method of monitoring context using the same and context monitoring system having the same (US8599710B2)

PENDING

Apparatus, method, and system for multi-device wearability-aware heat orchestration (US20250264933A1, CN120523674A, EP4607318A1, GB2638425A)

Method for multi-factor user authentication (US20250211586A1, EP4575849A1)

Machine learning model scheduler (US20250165295A1, EP4557096A1, GB2635567A)

Apparatus & method for multi-device authentication (US20250094559A1, EP4524786A1, GB2633624A)

Earphone Modes of Operation (US20240040299A1, EP4312436A1)

Object Identification (US20230394686A1, EP4290472A1)

Resource control (US20230269668A1, WO2022023618A1, EP4190042A4)

Resource assessment of sensors (US20220330896A1, CN114402179A, EP4031838A1, WO2021053444A1)

Model modification and deployment (US20210350280A1)

Honors & Awards

2024	Best Paper Award , ACM HumanSys	Hangzhou, China
2022	Best Paper Award , ACM AIChallengIoT	Boston, USA
2021	Best Student Paper Award , ICMU	Tokyo, Japan
2019	Honorable Mention Award , ACM MobileHCI	Taipei, Taiwan
2016	Outstanding Ph.D. Thesis Award , KAIST	Daejeon, South Korea
2014	Best Paper Award , ACM CSCW	Baltimore, MD, USA
2013	Qualcomm Fellowship Award , Qualcomm	Daejeon, South Korea
2013	Best Demo Award , ACM HotMobile	Georgia, USA

Tutorials

2019	Building Embedded AI Systems - A Practical Approach , ACM MobiCom	Los Cabos, Mexico
2018	Designing Connected Data Products , ACM UbiComp	Singapore, Singapore

Talks

AI-powered Evolution in Collaborative Wearables with Multi-device, Multi-modal Systems, CS Department Special Seminar, POSTECH, Pohang, South Korea, Aug. 2024

AI-powered Evolution in Wearables with Multi-device, Multi-modal, Collaborative Systems, Keynote Speech, NetAISys, Tokyo, Japan, Jun. 2024

What's Next for AI Systems? Invited Lecture, KAIST, Daejeon, South Korea, Dec. 2023

Building Collaborative AI Systems for Multi-device Environment, Guest Lecture, University of Southampton, Southampton, UK, Nov. 2022

Building AI Systems for Collaborative, Interactive, and Privacy-Preserving Applications, Special Seminar, POSTECH, Pohang, South Korea, Nov. 2022

Building Collaborative AI Systems for Multi-device Environments, CS Seminar, UNIST, Ulsan, South Korea, Nov. 2022

The Challenge and Future of Sensory AI Systems: Making On-device AI a Reality, Invited Talk, SIGMOBILE CEP, Online, Mar. 2022

Earable Computing for Personal-scale Behavioral Analytics, Invited Talk, Daejeon, South Korea, Jul. 2021

What is Next for Sensory AI Systems? A Journey toward Making AI Work in the Real World, Special Seminar, Yonsei University, Incheon, South Korea, Jun. 2021

What is Next for AI Systems? A Journey toward Making AI Work in the Real World, Special Seminar, POSTECH, Pohang, South Korea, May 2021

Sensory AI Software Platform for Multi-device Environments, Special Session Invited Talk, ICTC, Jeju Island, South Korea, Oct. 2020

Sensory AI Software Platform for Multi-device Environments, Industry Track Invited Talk, IEEE SMARTCOMP, Bologna, Italy, Sep. 2020

Making AI Work in Multi-device World, Guest Lecture, Seoul National University, Seoul, South Korea, Jun. 2019

Resource Orchestration Platform for Life-immersive Sensing Applications, Invited Talk, Future Internet Forum Workshop, Seoul, South Korea, Oct. 2016

Ecosystem-wide Support for Power Impact-Awareness for Mobile Sensing Applications, Invited Talk, Nokia Bell Labs, Antwerp, Belgium, Sep. 2016

Ecosystem-centric Power Management for Continuous Sensing Applications, Invited Talk, Naver Labs, Sungnam, South Korea, Feb. 2016

An Active Resource Use Orchestration Framework for Mobile Context Monitoring in Sensor-rich Mobile Environments, Invited Talk, Research Center for UX-oriented Software Platform, Daejeon, South Korea, Feb. 2013

CoMon: Cooperative Ambience Monitoring Platform with Continuity and Benefit Awareness, Invited Talk, POSTECH, Pohang, South Korea, Aug. 2012

Scholaly Services

EDITORSHIP

- 2023- **IEEE Pervasive Computing**, Editorial Board Member
- 2017- **ACM IMWUT**, Associate Editor

ORGANISING COMMITTEE

- | | | |
|------|---|-----------------------------|
| 2026 | ACM MobiSys , Vice General Chair | <i>Cambridge, UK</i> |
| 2026 | ACM Edge-Well , Organizing Committee | <i>Pisa, Italy</i> |
| 2025 | ACM UbiComp/ISWC , Sponsorship and Industrial Relations Chairs | <i>Helsinki, Finland</i> |
| 2024 | ACM MobiSys , Student Travel Grants Chair | <i>Tokyo, Japan</i> |
| 2023 | ACM IASA , Sponsorship Chair | <i>San Antonio, USA</i> |
| 2022 | ACM UbiComp/ISWC , Workshop Chair | <i>Cambridge, UK</i> |
| 2021 | ACM EarComp , Program Chair | <i>Online</i> |
| 2020 | ACM SenSys , Poster and Demo Chair | <i>Yokohama, Japan</i> |
| 2019 | ACM EarComp , Local Arrangement | <i>London, UK</i> |
| 2019 | ACM WearSys , Program Chair | <i>Seoul, South Korea</i> |
| 2019 | ACM MobiSys , Demo and Video Chair | <i>Seoul, South Korea</i> |
| 2018 | ACM UbiComp , Publication Chair | <i>Singapore, Singapore</i> |
| 2017 | IEEE MDM , Local Organising Chair | <i>Daejeon, South Korea</i> |

PROGRAM COMMITTEE

- 2026 **ACM SenSys**, **ACM MobiCom**,
- 2025 **ACM MobiSys**, **ACM SenSys**, **IEEE PerCom**, **ACM MobiCom**,
- 2024 **ACM MobiSys**, **ACM BodySys**, **ACM EdgeSys**,
- 2023 **ACM IASA**,
- 2022 **ACM MobiSys**, **ACM MobiCom**, **ACM CML-IoT**, **ACM DIGIBIOM**,
- 2021 **ACM HotMobile**, **COMSNETS**, **ICMU**, **MobiQuitous**, **ICMU**, **MFSens**,
- 2020 **ACM WearSys**, **WCNC**, **COMSNETS**,
- 2019 **ACM MobiSys**, **ACM EarComp**, **PervasiveHealth**, **MUSICAL**,
- 2017 **IEEE MDM**,
- 2015 **Embarrassing Interactions**, **ACM MobiSys PhD Forum**,